Predicting Full-Benchmark Scores from Few Items: Predictors Fail Under Shift, Corrected and Recalibrated Estimators Do Not

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Abstract

Evaluating a language model on every item of a large benchmark is costly, so many methods run the model on a small subset (a "coreset") and predict its full-benchmark score using item-level results of earlier models. We frame this as finite-population estimation, where a method is a *design* (which items) plus an *estimator*, and compare designs and estimators on 13 binary item-level benchmarks (Open LLM Leaderboard v2 and HELM) with 79–448 models, at coreset sizes 10–200, under three source/target relationships: random model splits, held-out organisations, and a deliberately demanding extrapolation where the strongest 25% of models are targets. We find (i) difficulty-stratified sampling, which needs no model, lowers RMSE by about 17% relative to random sampling in every setting; (ii) learned predictors (IRT imputation, ridge regression, anchor points with IRT) cut RMSE by roughly 20–50% (depending on the budget) for interpolation but become worse than random sampling under extrapolation (relative RMSE 1.4–1.65 at $n = 100$), through a systematic downward bias of about 5 accuracy points; (iii) estimators with a residual correction, or a simple per-target recalibration of the IRT logits (Cal-IRT), stay better than random sampling in all settings, with Cal-IRT 8% (95% CI 3–13%) lower RMSE than an AIPW baseline under extrapolation at $n = 100$ and a pretest-shrunk version (ShrinkCal-IRT) recovering most of the interpolation gain; (iv) a variance-aware unequal-probability design (VA-PPS) helps only marginally ($\leq$5–7%), and (v) choosing among methods by pseudo-shift validation within the source models did not give robustness. All intervals are hierarchical bootstrap intervals over benchmarks and seeds; code regenerates every number.

1 Introduction

Evaluating a language model on a benchmark with thousands of items is expensive when repeated for many checkpoints, prompts or decoding settings. A tempting remedy is to run the new ("target") model on a small subset and extrapolate to the full benchmark, using the item-level records of earlier ("source") models to choose the subset and to calibrate the extrapolation. Many methods follow this recipe—clustering-based anchor points, IRT-based imputation, learned assessors, active item selection [Vivek et al., 2024, Maia Polo et al., 2024, Kipnis et al., 2025, Pacchiardi et al., 2024, Li et al., 2025]—and report accurate predictions from a small fraction of the items. A recent large comparison cautions that these gains are largest when the target resembles the source models, and shrink or vanish otherwise [Zhang et al., 2025]. Whether to trust a predicted score therefore depends on how the target relates to the source population, which is exactly what is unknown for a new model.

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We study this question with a simple frame: the benchmark is a finite population of items, the target's accuracy is a population mean, the coreset is a sample, and source models supply auxiliary information. In this frame a method has a *design* (which items) and an *estimator* (how to turn the target's coreset responses into a score), and the two can be varied independently; the classical survey-sampling toolkit—stratification, Horvitz–Thompson weights, Neyman allocation, model-assisted residual correction—gives a vocabulary and a set of strong "simple" baselines that predictor-based work often omits.

Contributions. (i) A controlled comparison on 13 binary item-level benchmarks (7 Open LLM Leaderboard, 6 HELM) of designs and estimators, under three kinds of source/target relationship (random split, held-out organisations, stronger targets) at five budgets, with paired hierarchical-bootstrap uncertainty. (ii) Two small ingredients: *VA-PPS*, a variance-aware unequal-probability design for residual-corrected estimators, and *Cal-IRT* / *ShrinkCal-IRT*, per-target logit recalibration of an IRT predictor (with a pretest shrinkage between uncalibrated and recalibrated predictions). (iii) A negative result for tuning by pseudo-shift validation within the source models. (iv) Complete code to regenerate every table and figure.

2 Related work

Benchmark prediction from few items. Anchor Points [Vivek et al., 2024] cluster items by the response vectors of source models and evaluate a new model on one representative per cluster, weighting by cluster size. tinyBenchmarks [Maia Polo et al., 2024] combine such anchors with a multidimensional item-response-theory (IRT) model to impute the unseen items; metabench [Kipnis et al., 2025] selects items by IRT information and predicts with generalised additive models. Pacchiardi et al. [2024] train assessor models on a few reference instances, and Li et al. [2025] learn an adaptive item-acquisition policy. Perlitz et al. [2024] study the same cost problem from a benchmark-reduction view, and Rodriguez et al. [2021] use IRT to show that items differ widely in informativeness. Zhang et al. [2025] compare eleven such methods on 19 benchmarks and report that most of them interpolate well among similar models but struggle when the target differs from the models seen before, whereas random sampling and an AIPW estimator stay reliable (we know this work only through its description in the task's starter list and the README of its data repository; see Appendix F). Our experiments use the item-level data released with that work, but our protocols, estimators and designs are implemented independently.

Survey sampling and prediction-powered inference. Estimating a finite-population mean from a sample is a classical problem. Horvitz–Thompson weighting [Horvitz and Thompson, 1952] gives design-unbiased estimates under unequal-probability sampling, Neyman allocation [Neyman, 1934] minimises variance given stratum variances, and model-assisted (GREG/difference) estimators [Särndal et al., 1992] add a design-weighted residual correction to model predictions, so that the estimate remains unbiased however wrong the model is. The same construction underlies doubly-robust/AIPW estimation [Robins et al., 1994] and prediction-powered inference [Angelopoulos et al., 2023]. We treat benchmark prediction as a finite-population problem where the items are the population, the (unknown) target model's correctness is the variable of interest, and source models provide the auxiliary information. This view makes explicit the trade-off studied here: pure predictors are efficient but can be biased under distribution shift, design-based estimators are unbiased but only modestly efficient.

IRT for model evaluation and recalibration. Multidimensional IRT is standard in psychometrics and was adopted for LLM evaluation by several of the works above. Recalibrating predicted logits with a fitted slope and intercept goes back to Platt scaling [Platt, 1999]; we apply it per target model, to the IRT logits.

3 Problem setting and methods

Setting. A benchmark has N items. We have a binary source matrix $Y^S \in \{0,1\}^{m\times N}$ (m source models). A target model t has an unknown full-benchmark accuracy $a_t = \frac{1}{N}\sum_{i=1}^{N} y_{ti}$. A *coreset* $C \subset [N]$ with $|C| = n \ll N$ is chosen using Y^S only; we then observe y_{tC} and output $\hat{a}_t$. A method

is thus a pair (*design*, *estimator*). The same coreset is used for all targets in a split (as in practice, where one subset is run for every checkpoint). Error is measured against the true a_t.

3.1 Designs (how to choose C)

Each design returns C and population weights w_i with $\sum_{i \in C} w_i \approx 1$; for inclusion probability π_i, $w_i = 1/(N\pi_i)$. **Random**: simple random sample without replacement. **Difficulty-stratified**: sort items by source-model accuracy $\bar{p}_i = \frac{1}{m}\sum_j y_{ji}$ and take a systematic sample of size n along this order (equal $\pi_i = n/N$; implicit stratification). **Anchor points** [Vivek et al., 2024]: k-means with $k = n$ on the item response vectors $Y_{\cdot i}^S$, one medoid per cluster, weighted by cluster size$/N$. **D-optimal (information-based)**: greedily maximise $\log\det$ of the Fisher information of the fitted IRT model, a stand-in for IRT-information selection [Kipnis et al., 2025]. **VA-PPS** (variance-aware PPS; ours): $\pi_i = \min\{1, nq_i\}$ (water-filled so that $\sum_i \pi_i = n$) with $q_i \propto (1-\lambda)/N + \lambda\, s_i / \sum_k s_k$, $s_i = \sqrt{v_i}$, where v_i is the mean squared residual of the fitted source IRT model on item i (items on which source models are predictable carry little residual variance). Items are drawn by fixed-size systematic sampling along the difficulty order using π_i, which gives exactly n items, stratification by difficulty, and known π_i. We use $\lambda = 1$ (pure Neyman-type allocation) as the main variant and $\lambda = 0.5$ (uniform floor) in ablations.

Proposition 1 (Neyman allocation for residual correction)**.** *Let $r_i = y_{ti} - f_i$ be the residuals of any fixed predictor f, and let item i be included independently with probability π_i (Poisson sampling). Then $\hat{a} = \frac{1}{N}\sum_i f_i + \frac{1}{N}\sum_{i \in C} r_i/\pi_i$ is unbiased for a_t with variance $N^{-2}\sum_i(\pi_i^{-1} - 1)r_i^2$. Under $\sum_i \pi_i = n$ and $\pi_i \le 1$, this is minimised by $\pi_i \propto |r_i|$ (capped at 1).*

Proof. Unbiasedness: $\mathbb{E}[\mathbb{1}_{i \in C}/\pi_i] = 1$. Variance: independent Bernoulli inclusions give $\mathrm{Var}(\mathbb{1}_i r_i/\pi_i) = r_i^2(1-\pi_i)/\pi_i$. Minimising $\sum_i r_i^2/\pi_i$ s.t. $\sum_i \pi_i = n$ by Lagrange gives $\pi_i \propto |r_i|$; items hitting the cap $\pi_i = 1$ are fixed and the remainder re-scaled. □

Since r_i is unknown for the target, VA-PPS uses $\sqrt{v_i}$, an estimate of $\sqrt{\mathbb{E}_j r_{ji}^2}$ from source models; this is only a proxy and is not variance-optimal for a target that differs from the sources (the uniform floor $\lambda < 1$ protects against $v_i \approx 0$ items on which a shifted target errs). The systematic (rather than Poisson) draw yields fixed n; Proposition 1 then holds only approximately, and we evaluate the design empirically.

3.2 Estimators (how to predict a_t)

Let $W = \sum_{i \in C} w_i$. **Mean**: $\hat{a} = \sum_{i \in C} w_i y_{ti}/W$ (the Hajek mean; the plain accuracy on the coreset for equal weights). **Linear calibration**: regress source full accuracies on source coreset accuracies, apply to the target's coreset accuracy. **Ridge**: ridge regression (RidgeCV, penalty chosen by efficient leave-one-out on the source models) from a source model's coreset responses $y_{jC} \in \{0,1\}^n$ to its full accuracy; a representative of the "learn a map from the coreset to the score" family (cf. Lasso/PCA-type predictors). **IRT family.** We fit a multidimensional 2PL model $P(y_{ji} = 1) = \sigma(\theta_j^\top \alpha_i + \beta_i)$ with $\theta_j, \alpha_i \in \mathbb{R}^d$ on Y^S by L-BFGS with ℓ_2 penalties (details in Appendix B); for a target we compute the MAP ability $\hat{\theta}_t$ from y_{tC} (prior $\mathcal{N}(0, 4I)$) and $z_{ti} = \hat{\theta}_t^\top \alpha_i + \beta_i$, $\hat{p}_{ti} = \sigma(z_{ti})$. Then: *IRT (pure)*: $\hat{a} = \frac{1}{N}\sum_i \hat{p}_{ti}$. *Plug-in* (tinyBenchmarks-style imputation [Maia Polo et al., 2024]): $\hat{a} = \frac{1}{N}(\sum_{i \in C} y_{ti} + \sum_{i \notin C} \hat{p}_{ti})$. *IRT-AIPW*: $\hat{a} = \frac{1}{N}\sum_i \hat{p}_{ti} + \sum_{i \in C} w_i(y_{ti} - \hat{p}_{ti})$, the model-assisted/AIPW form [Särndal et al., 1992, Robins et al., 1994, Angelopoulos et al., 2023]. *Cal-IRT* (ours): per-target recalibration of the logit, $\tilde{p}_{ti} = \sigma(s_t z_{ti} + c_t)$ with (s_t, c_t) maximising the coreset log-likelihood with a small ridge penalty $\frac{1}{2}((s-1)^2 + c^2)$ towards $(1, 0)$ (Platt-style [Platt, 1999]); $\hat{a} = \frac{1}{N}\sum_i \tilde{p}_{ti}$. The intercept c_t absorbs a global shift in difficulty between the target and the source population, which is what "stronger target models" produce. It costs two parameters estimated from n responses. *ShrinkCal-IRT* (ours): a pretest/positive-part shrinkage between pure and recalibrated predictions. With $\bar{r}_t = \sum_{i \in C} w_i(y_{ti} - \hat{p}_{ti})$ and $\hat{v}_t$ the estimated variance of $\bar{r}_t$ (sample variance of coreset residuals$/n$, finite-population corrected), $\kappa_t = \max\{0, 1 - c_0 \hat{v}_t/\bar{r}_t^2\}$ and $\hat{a} = \mathrm{pure} + \kappa_t(\mathrm{Cal} - \mathrm{pure})$, $c_0 = 0.5$. A residual indistinguishable from noise leaves the pure prediction untouched; a clearly non-zero residual switches the recalibration on. We also report cross-fitted AIPW (*gregcf*, residuals from a 5-fold cross-fitted ability) and Cal-IRT with the AIPW correction (*calgreg*).

3.3 Pseudo-shift tuning (AUTO)

Because the best estimator depends on an unknown shift, we also evaluate a selector that uses only source models. The source set is split twice: a random 30% and the strongest 25% of source models serve as pseudo-targets. For each candidate (design $\in$ {stratified, VA-PPS $\lambda \in \{0.5, 1\}$} $\times$ estimator $\in$ {mean, linear, ridge, and pure/plug-in/AIPW/Cal/Cal-AIPW/ShrinkCal IRT with $d \in \{2, 4, 8\}$}) and budget, we compute RMSE relative to difficulty-stratified sampling on the pseudo-targets (2 repetitions each); *minimax* picks the candidate with the lowest worst-case ratio over the two inner splits and *mean* the lowest average ratio. The chosen candidate is then applied to the real targets.

4 Experimental protocol

Data. We use the binary item-level matrices released with Zhang et al. [2025] (fixed commit `55faff4`, MIT licence), derived from the Open LLM Leaderboard v2 [Fourrier et al., 2024] and HELM [Liang et al., 2023]: seven Open LLM Leaderboard benchmarks (ARC-Challenge, BBH, GPQA, IFEval, MMLU-Pro, MuSR, MATH Lvl 5; 218–448 models, 541–12 032 items) and six HELM scenarios (files named commonsense, gsm, legalbench, math, med_qa, mmlu; 79–83 models, 437–2 047 items). Model sets differ per benchmark (Appendix Table 3). We did not re-run any model.

Splits (who is the target?). For each benchmark and seed we split models into source and target sets in one of three ways. **Random**: 30% of models are targets, drawn uniformly (the interpolation setting most prior work evaluates). **Held-out organisations**: whole organisations (the prefix of the model id; Open LLM Leaderboard only, since HELM model ids are not aligned with all matrices) are moved to the target set until $\geq 25\%$ of the models are targets; this removes near-duplicate fine-tunes and merges of the same base model from the sources. **Stronger targets**: the 25% of models with the highest full-benchmark accuracy are targets, all others are sources. This is a deliberately demanding but realistic extrapolation (a new, better model is evaluated using the records of older, weaker ones); the split itself is deterministic and seeds vary the coreset and model initialisation. Extra severities (top 10%, top 50%, weakest 25%) are in Appendix D.

Budgets, seeds, metrics. Coreset sizes $n \in \{10, 25, 50, 100, 200\}$; 30 seeds per (benchmark, split type) (random splits differ per seed); all methods share the same split within a seed. For each run we compute, over the target models of the split, RMSE, MAE and signed mean error of $\hat{a}_t - a_t$ (accuracy in $[0, 1]$), and Spearman's ρ and Kendall's τ between $\hat{a}_t$ and a_t (note that with a shared coreset the common component of the error cancels in rankings). To compare across benchmarks with very different scales (e.g. MATH Lvl 5 accuracies are around 5%) we report *relative RMSE*: for a benchmark, split type and n, $\rho_b = \sqrt{\overline{\mathrm{MSE}}_{\text{method}} / \overline{\mathrm{MSE}}_{\text{random sampling}}}$ with MSE averaged over seeds, aggregated across benchmarks by the geometric mean (below 1 = better than random sampling with the same n). 95% confidence intervals come from a hierarchical bootstrap that resamples benchmarks and then seeds within benchmarks (1000 resamples), jointly for all methods (so differences between methods are paired). Intervals reflect variability over benchmarks, splits and coresets; with the deterministic stronger-targets split they do *not* reflect variability in which models happen to be the strongest.

Baselines and hyper-parameters. Baselines are random sampling, difficulty-stratified sampling, linear calibration, ridge regression, Anchor Points, anchors with IRT plug-in (tinyBenchmarks-style), D-optimal items with IRT plug-in (information-based), and IRT-AIPW with uniform sampling (the AIPW-style estimator that Zhang et al. [2025] recommend; our implementation, not theirs). The IRT dimension is fixed to $d = 4$ in the main tables ($d \in \{2, 4, 8\}$ in ablations). The IRT regularisers, the ability prior variance (4), the Platt penalty (1) and $c_0 = 0.5$ were fixed using a separate development run (seeds ≥ 1000 and 2000–2007, on 9–11 benchmarks, budgets 25 and 100; the development benchmarks did not include BBH or MMLU-Pro) and never tuned on the evaluation seeds. The main variants (VA-PPS $\lambda = 1$ with Cal-IRT and with ShrinkCal-IRT, $d = 4$) were declared before the main run; the many other combinations in the ablations are reported for completeness and not selected post hoc.

Table 1: Relative RMSE vs. random sampling (geometric mean over benchmarks; 95% CI). Rows above the rule are baselines; "ours" are the components proposed here; AUTO is source-only pseudo-shift tuning. Best entries per column in the interpolation settings are the learned predictors; in the extrapolation setting they are the recalibrated/residual-corrected estimators. The held-out-organisation column covers 7 benchmarks, the others 13. Full results for all five budgets are in Appendix Table 5.

	Random split		Held-out organisations		Stronger targets (extrapolation)	
	n=25	n=100	n=25	n=100	n=25	n=100
Random sampling	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Difficulty-stratified sampling	0.83 (0.79–0.88)	0.83 (0.79–0.87)	0.83 (0.76–0.89)	0.86 (0.80–0.91)	0.82 (0.78–0.87)	0.82 (0.78–0.87)
Random + linear calibration	0.67 (0.57–0.76)	0.79 (0.74–0.83)	0.61 (0.49–0.75)	0.77 (0.69–0.85)	0.95 (0.85–1.07)	1.07 (0.95–1.19)
Random + ridge regression	0.61 (0.50–0.72)	0.73 (0.63–0.83)	0.53 (0.41–0.67)	0.67 (0.56–0.81)	1.12 (0.94–1.32)	1.65 (1.37–1.96)
Anchor points (weighted mean)	0.78 (0.68–0.89)	1.08 (0.91–1.28)	0.82 (0.67–1.03)	1.21 (0.95–1.63)	0.90 (0.65–1.30)	1.34 (0.90–2.07)
Anchor points + IRT (tinyBenchmarks-style)	0.58 (0.47–0.71)	0.83 (0.66–1.07)	0.48 (0.40–0.58)	0.67 (0.58–0.81)	1.02 (0.85–1.22)	1.40 (1.11–1.73)
D-optimal items + IRT (info-based)	0.70 (0.55–0.92)	1.01 (0.79–1.33)	0.53 (0.41–0.67)	0.77 (0.63–0.96)	1.01 (0.85–1.20)	1.62 (1.33–1.98)
Random + IRT (pure)	0.65 (0.55–0.75)	0.78 (0.70–0.86)	0.57 (0.46–0.69)	0.71 (0.64–0.80)	1.08 (0.90–1.29)	1.54 (1.35–1.79)
Random + IRT-AIPW	0.78 (0.73–0.82)	0.80 (0.77–0.83)	0.77 (0.71–0.83)	0.80 (0.74–0.86)	0.78 (0.73–0.84)	0.82 (0.77–0.87)
VA-PPS + IRT-AIPW	0.78 (0.72–0.87)	0.82 (0.77–0.89)	0.75 (0.68–0.83)	0.77 (0.71–0.82)	0.76 (0.69–0.84)	0.88 (0.78–0.98)
VA-PPS + IRT (pure)	0.65 (0.54–0.79)	0.82 (0.69–1.00)	0.54 (0.45–0.64)	0.69 (0.64–0.76)	1.02 (0.86–1.20)	1.51 (1.30–1.76)
VA-PPS + Cal-IRT (ours)	0.69 (0.63–0.75)	0.75 (0.71–0.80)	0.65 (0.59–0.72)	0.74 (0.69–0.80)	0.70 (0.64–0.76)	0.76 (0.70–0.82)
VA-PPS + ShrinkCal-IRT (ours)	0.64 (0.58–0.72)	0.71 (0.66–0.77)	0.58 (0.51–0.65)	0.68 (0.62–0.74)	0.80 (0.71–0.90)	0.88 (0.82–0.93)
AUTO (pseudo-shift, minimax)	0.68 (0.59–0.76)	0.75 (0.71–0.80)	0.62 (0.51–0.73)	0.73 (0.67–0.78)	0.77 (0.68–0.87)	0.82 (0.74–0.91)
AUTO (pseudo-shift, mean)	0.66 (0.55–0.76)	0.72 (0.67–0.77)	0.58 (0.45–0.72)	0.70 (0.63–0.77)	0.81 (0.71–0.95)	0.87 (0.76–0.99)

5 Results

All numbers are relative RMSE against random sampling with the same n (below 1 is better), geometric mean over benchmarks, with 95% hierarchical-bootstrap intervals (Section 4); 13 benchmarks for random and stronger-target splits and the 7 Open LLM Leaderboard benchmarks for held-out organisations. For scale, random sampling with $n = 100$ has an RMSE of about 3–5 accuracy points on most benchmarks (Appendix Table 4); much of this is a component shared by all targets of a split, because they use one coreset.

Interpolation: learned predictors help a lot. When targets are drawn at random from the same population as the sources, predictors beat random sampling by wide margins at small budgets: with $n = 25$, ridge regression on the coreset responses reaches 0.61 (0.50–0.72), anchors with IRT imputation 0.58 (0.47–0.71), and pure IRT 0.65 (0.55–0.75). Gains shrink as n grows (0.73–0.83 at $n = 100$), as expected when the baseline error falls like $n^{-1/2}$. With held-out organisations the picture is the same or slightly stronger (anchors+IRT 0.48 at $n = 25$). Rankings improve too: Spearman's ρ rises by $+0.05$ to $+0.13$ at $n = 25$ (Appendix Table 6). Difficulty-stratified sampling, which needs no model at all, lowers RMSE by about 17% (0.83, CI 0.79–0.88) at every budget and in every split type, and wins on all benchmarks (Appendix Table 10).

Extrapolation: predictors become worse than random sampling. When the targets are the strongest quarter of the models, every method that predicts from the source population without a correction degrades. At $n = 100$, pure IRT has relative RMSE 1.54 (1.35–1.79), ridge 1.65 (1.37–1.96), anchors+IRT 1.40 (1.11–1.73) and D-optimal items+IRT 1.62 (1.33–1.98); they beat random sampling on only 1–3 of 13 benchmarks (Appendix Table 10). The error is systematic: Table 2 shows a mean signed error of -5 accuracy points, accounting for about three quarters of the MSE. Rank correlation also falls below that of random sampling (-0.08 to -0.16 in ρ; Appendix Table 6). The anchor-weighted mean, which is design-based but uses cluster sizes from the source population, is also no better than random sampling at $n = 100$ (1.34, CI 0.90–2.07) in this setting.

Residual correction or recalibration restores robustness. Model-assisted (AIPW) estimators remain design-unbiased under any shift: IRT-AIPW with uniform sampling has 0.78 (0.73–0.84) at $n = 25$ and 0.82 (0.77–0.87) at $n = 100$ under extrapolation and beats random sampling on all 13 benchmarks at both budgets. Cal-IRT with VA-PPS improves on this: 0.70 (0.64–0.76) and 0.76 (0.70–0.82), 13/13 wins, and is better than IRT-AIPW in paired comparisons by 11% and 8% (ratio 0.89 and 0.92 with intervals excluding 1; Appendix Table 8). Under interpolation, Cal-IRT keeps most of the gain (0.69 and 0.75 for random splits), but remains behind the best uncorrected predictors at small n (paired ratio vs. ridge 1.14 at $n = 25$, CI 1.00–1.34). Its bias under extrapolation is -0.5 points versus -5.2 for the uncalibrated IRT with the same design (Table 2). ShrinkCal-IRT trades

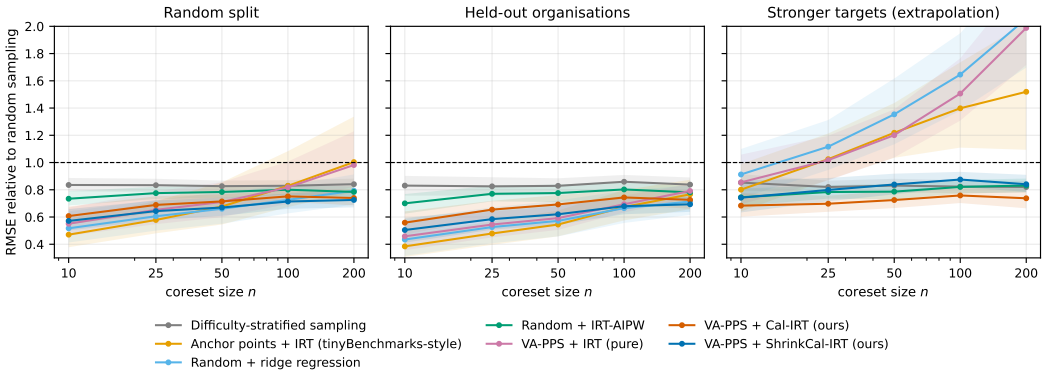


Figure 1: Relative RMSE vs. random sampling as a function of coreset size, per split type (shaded: 95% CI). The vertical axis is clipped at 2.0; pure-prediction methods continue upwards in the right panel.

Table 2: Signed error and its share in the MSE at $n = 100$: average over benchmarks of the mean signed error $\hat{a}_t - a_t$ over target models (accuracy points) and bias^2/MSE pooled over runs. A shared coreset produces some common-mode error even for unbiased methods.

	Random split		Held-out organisations		Stronger targets (extrapolation)	
	mean err (pts)	bias^2/MSE	mean err (pts)	bias^2/MSE	mean err (pts)	bias^2/MSE
Random sampling	-0.03	28%	-0.08	24%	-0.09	42%
Difficulty-stratified sampling	-0.00	4%	+0.05	3%	-0.13	18%
Random + linear calibration	+0.02	3%	-0.02	3%	-2.61	64%
Random + ridge regression	+0.11	4%	-0.05	3%	-5.66	76%
Anchor points (weighted mean)	-0.35	25%	-1.79	26%	-1.64	25%
Anchor points + IRT (tinyBenchmarks-style)	+0.38	8%	-0.20	5%	-5.04	73%
D-optimal items + IRT (info-based)	+0.65	11%	-0.44	11%	-5.83	77%
Random + IRT (pure)	-0.03	4%	-0.17	5%	-5.23	75%
Random + IRT-AIPW	-0.01	4%	+0.02	2%	-0.31	18%
VA-PPS + IRT-AIPW	+0.03	3%	-0.04	2%	-0.07	14%
VA-PPS + IRT (pure)	+0.17	6%	-0.16	4%	-5.15	75%
VA-PPS + Cal-IRT (ours)	-0.04	3%	-0.08	2%	-0.48	18%
VA-PPS + ShrinkCal-IRT (ours)	-0.02	3%	-0.11	3%	-1.50	31%
AUTO (pseudo-shift, minimax)	+0.01	4%	-0.10	3%	-1.32	40%
AUTO (pseudo-shift, mean)	-0.05	3%	-0.08	3%	-1.69	45%

in the other direction: it recovers most of the interpolation advantage (0.64/0.71 for random splits, within the intervals of ridge) and costs 15% relative to Cal-IRT under extrapolation (0.80/0.88, still better than random sampling on 13/13). No method dominates: the choice between Cal-IRT and ShrinkCal-IRT is a choice about how much extrapolation to expect.

Designs matter much less than estimators. Holding the estimator fixed, difficulty stratification changes RMSE by $\leq 4\%$ for IRT-based estimators, and VA-PPS adds at most 5–7% over stratification, significantly so only in some columns (Appendix Table 9; e.g. Cal-IRT, VA-PPS vs. stratified: 0.96 at $n = 25$ and 0.95 at $n = 100$ with held-out organisations, 0.93 at $n = 25$ under extrapolation, but 0.98–0.99 elsewhere with intervals covering 1). The pure Neyman allocation ($\lambda = 1$) is a poor design for estimators that use the Horvitz–Thompson weights alone: the weighted mean with VA-PPS $\lambda = 1$ is 9–47% worse than stratified sampling, and VA-PPS $\lambda = 1$ with IRT-AIPW is slightly worse than with $\lambda = 0.5$ (0.88 vs 0.78 at $n = 100$ under extrapolation; Appendix Table 11). Information-based (D-optimal) item selection, which concentrates items in a few directions of latent space, is worse than random or stratified items for the same estimators in our setting. The IRT dimension ($d \in \{2, 4, 8\}$) matters little for the corrected estimators; larger d is slightly worse under extrapolation.

Pseudo-shift tuning does not buy robustness. Tuning the (design, estimator) pair on pseudo-shifts of the source models (AUTO) does reasonably in the interpolation settings (e.g. 0.68/0.75 at $n = 25/100$ for random splits, minimax) but is 8–11% worse than fixed Cal-IRT under extrapolation (paired ratio 1.11, CI 1.04–1.18, at $n = 25$; 1.08, CI 1.02–1.15, at $n = 100$), and 17–31% of its minimax choices there are uncorrected predictors (pure IRT, plug-in or ridge; Appendix Table 14),

which fail on the real shift. Holding out the top quarter of the *sources* produces a milder gap than the real extrapolation, so the inner validation under-estimates the bias. We view this as a caution against validating shift-robustness only by within-source splits.

Shift severity. In a smaller additional run (10 seeds, $n \in \{25, 100\}$, four designs; Appendix D) we varied which models are targets: the strongest 10% or 50% of models, or the weakest 25%. Uncorrected predictors fail in every direction of extrapolation, and most when the sources are only the weaker half: at $n = 100$ pure IRT is 1.16 (1.04–1.36) of random-sampling RMSE for the strongest-10% targets and 2.07 (1.77–2.49) for strongest-50%, and for the weakest-25% targets they over-predict by 8–9 points (relative RMSE 2.1). Cal-IRT is below 1 in all four settings at both budgets, but for the weakest-25% targets its intervals include 1 (0.90, 0.77–1.06 at $n = 25$; 0.92, 0.81–1.05 at $n = 100$) and so does IRT-AIPW's, so we cannot claim a gain over random sampling there; difficulty-stratified sampling (0.92 and 0.90) is the only method whose interval stays at or below 1.

Number of source models. In a further run on 7 benchmarks (10 seeds, $n \in \{25, 100\}$, stratified and VA-PPS designs) we kept only $m \in \{20, 50, 100\}$ randomly chosen source models (Appendix E). With $m = 20$, all model-based methods are clearly weaker, and under extrapolation difficulty-stratified sampling (0.79 at $n = 25$, 0.83 at $n = 100$) is better than all of them (Cal-IRT 0.86 and 0.94, intervals covering 1 at $n = 100$); with $m = 50$ Cal-IRT is already at 0.76 / 0.85 and with all sources 0.68 / 0.73 under extrapolation. Interpolation gains of uncorrected predictors also need $m \gtrsim 50$ (ridge 0.71 / 0.56 / 0.52 at $n = 25$ for $m = 20/50/100$). The same figures are in a table in the appendix; note that for the HELM benchmarks the number of available source models is below 100, so $m = 100$ equals the full source set there.

6 Discussion and limitations

What the evidence supports. On these 13 binary benchmarks, (i) a model-free change of design (difficulty stratification using source records) gives a consistent $\sim$17% RMSE reduction; (ii) learned predictors give large further gains when the target is an interpolation of the source population, and become worse than random sampling—through a systematic downward bias—when the targets are stronger than every source model; (iii) per-target recalibration of an IRT predictor (Cal-IRT) or its pretest-shrunk version (ShrinkCal-IRT) retains much of the interpolation gain while staying better than random sampling in all our extrapolation settings, and Cal-IRT beats the AIPW estimator by 6–15% in paired comparisons; (iv) the contribution of the variance-aware design is small ($\leq$5–7%, not always distinguishable from zero), and the uniform floor ($\lambda < 1$) is safer; (v) selecting among methods by pseudo-shift validation inside the source set was not effective. In sample-size terms, and assuming the error of random sampling scales as $n^{-1/2}$, a relative RMSE of 0.70–0.76 corresponds to the accuracy of $\approx$1.7–2$\times$ as many random items. Practical reading: if one cannot rule out that a new model lies outside the span of the source models—the usual case for a new, stronger model—an uncorrected predictor can be worse than simply running random items, and a corrected or recalibrated estimator on a difficulty-stratified or VA-PPS coreset is the safer default; if the target is a sibling of known models, uncorrected predictors (or ShrinkCal-IRT) are better.

Limitations. *Scope.* Only binary correctness on 13 benchmark matrices from two sources was studied; continuous or open-ended scores, chat/judge benchmarks and benchmarks with very few source models are not covered. The HELM benchmarks have only 79–83 models (about 20 targets per split), so their results are noisier. *Shift is operationalised, not measured.* "Stronger targets" (top 25% by accuracy) is one particular, fairly severe extrapolation; the ordering of methods under milder or different shifts (other capability profiles, different eras of models, new architectures) may differ, and the split is deterministic, so the intervals do not account for which models are strongest. Leaderboard models include many fine-tunes and merges of a few base models, which flatters interpolation (organisation hold-out only partly addresses this). *Target accuracy is on the benchmark itself.* All targets are scored on the same finite item set; we say nothing about generalisation to unseen items, nor about evaluation noise in the underlying leaderboards (e.g. sampling temperature, prompt format). *Baselines are re-implementations.* Anchor Points, tinyBenchmarks-style, D-optimal/metabench-style and AIPW baselines are our own simplified versions on binary data (e.g. k-means on binary vectors instead of confidence-based clustering, a fixed IRT dimension, no cross-validated mixing weights);

they are not the original authors' code, and stronger tuned versions may close some gaps. *Tuning.* IRT regularisation, the ability prior, the Platt penalty and c_0 were set on a separate development run using 9–11 of the benchmarks and fixed values; $c_0 = 0.5$ was not varied in the main run, and the VA-PPS λ was fixed to the value that looked best for Cal-IRT in development, which turned out to be a worse choice than $\lambda = 0.5$ for AIPW-type estimators. *No guarantees.* Cal-IRT and ShrinkCal-IRT are heuristics: they are not unbiased (Cal-IRT has a small negative bias under extrapolation, -0.5 points), the shrinkage is a pretest rule without formal risk guarantees, and Proposition 1 concerns Poisson sampling and a fixed predictor while we use systematic sampling with a proxy variance. *Metric.* Relative RMSE averages errors over target models sharing a single coreset; ranking metrics are computed within a split and are not comparable across split types. *Cost.* IRT fitting on the largest benchmark takes on the order of 10–30 s per source set on one CPU core; per-target inference is negligible.

Broader impact. Cheaper evaluation lowers costs, but over-trusting a predicted score (particularly for new, stronger models) can mislead model comparison; we recommend reporting the coreset size and an uncertainty interval, and not using predicted scores for decisions near ties.

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A Data

Table 3: Item-level matrices used (OLL: Open LLM Leaderboard v2; HELM). "const. items": items answered identically by all models. Accuracy statistics are over models.

benchmark	source	models	items	mean acc.	sd acc.	range	const. items
arc challenge	OLL	218	1172	52.2	12.9	20.5–75.1	32
bbh	OLL	446	5761	44.7	11.6	24.1–74.7	0
gpqa	OLL	447	1192	28.9	3.8	22.5–43.7	0
ifeval	OLL	448	541	33.3	20.4	3.7–88.0	7
mmlu pro	OLL	448	12032	28.1	12.9	10.3–59.7	1
musr	OLL	448	756	39.9	4.1	32.3–53.6	10
openllm math	OLL	447	894	5.1	6.8	0.0–35.6	119
commonsense	HELM	83	500	82.3	17.3	22.2–97.2	0
gsm	HELM	83	1000	65.7	24.9	2.8–95.6	3
legalbench	HELM	79	2047	53.3	10.6	22.8–70.2	53
math	HELM	83	437	56.8	27.4	2.5–93.4	1
med qa	HELM	83	1000	59.4	16.4	21.5–86.9	2
mmlu	HELM	83	567	61.3	12.4	22.9–81.1	1

Table 4: Absolute RMSE (accuracy points; $\sqrt{\text{seed-averaged MSE}}$) at $n = 100$ for random splits and stronger targets.

split	method	arc challenge	bbh	commonsense	gpqa	gsm	ifeval	legalbench	math	med qa	mmlu	mmlu pro	musr	openllm math
random	Random sampling	4.4	4.6	3.1	4.2	3.9	3.8	4.8	3.5	4.7	4.0	4.1	4.1	1.9
random	Difficulty-stratified sampling	3.3	4.1	2.4	4.1	3.3	3.2	3.5	3.0	3.7	3.0	3.5	3.6	1.7
random	Random + IRT-AIPW	3.1	3.7	2.7	3.7	3.2	3.0	3.4	3.0	3.5	3.1	3.3	3.2	1.7
random	VA-PPS + Cal-IRT (ours)	2.8	3.5	2.5	3.5	3.1	2.9	3.3	3.2	3.5	2.9	3.1	2.8	1.3
random	VA-PPS + ShrinkCal-IRT (ours)	2.5	3.0	2.6	3.0	3.3	2.8	3.3	3.3	3.5	2.9	2.8	2.4	1.3
top	Random sampling	4.4	4.6	1.9	4.5	2.7	4.0	5.0	3.0	4.3	3.9	4.5	4.1	3.3
top	Difficulty-stratified sampling	3.2	4.0	1.6	3.9	2.4	3.7	3.4	2.4	3.0	2.9	4.0	3.8	3.0
top	Random + IRT-AIPW	2.9	4.0	1.8	4.1	2.3	3.5	3.3	2.7	3.2	2.8	4.1	3.2	3.0
top	VA-PPS + Cal-IRT (ours)	2.8	3.6	1.3	3.8	2.3	3.5	2.9	2.3	3.4	2.5	4.1	3.0	2.8
top	VA-PPS + ShrinkCal-IRT (ours)	3.2	4.0	1.8	4.1	2.7	3.8	3.4	2.8	3.9	3.1	4.4	3.3	3.2

B Implementation details

IRT. $P(y_{ji} = 1) = \sigma(\theta_j^\top \alpha_i + \beta_i)$, minimised over (θ, α, β) with scipy L-BFGS-B (max. 150 iterations, 20 correction pairs, float32 matrix products) on the sum of binary cross-entropies plus $\frac{1}{2}\|\theta\|^2 + \frac{1}{2}\|\alpha\|^2 + 0.05\|\beta - \beta^0\|^2$, where $\beta_i^0 = \text{logit}(\text{clip}(\bar{p}_i, 0.02, 0.98))$ and initial $\theta, \alpha \sim 0.1\mathcal{N}(0, I)$. Target ability: MAP with prior $\mathcal{N}(0, 4I)$ by batched damped Newton iterations (step clipped to ± 2, at most 25 iterations). Platt recalibration: two-parameter Newton (at most 20 iterations, steps clipped to ± 1) on the coreset, ridge penalty 1 towards $(s, c) = (1, 0)$. ShrinkCal: $\hat{v}_t = \text{Var}_C(y - \hat{p})/n \cdot (1 - n/N)$ (unbiased sample variance), $c_0 = 0.5$.

VA-PPS. v_i is the mean squared residual of the $d = 4$ IRT fit on the source models; $s_i = \sqrt{\max(v_i, 10^{-4})}$; inclusion probabilities by water-filling to $\sum \pi_i = n$; systematic sampling over difficulty-sorted items (ties broken at random) with a uniformly random start.

Anchor points. scikit-learn `KMeans` with $k = n$, `n_init`=1, `max_iter`=30 on the source item vectors; the item closest to each centroid is the anchor, weight = cluster size$/N$.

D-optimal. Greedy $\log\det$ maximisation of $I+\sum_i \bar{w}_i \alpha_i \alpha_i^\top$ with $\bar{w}_i = \overline{p(1-p)}$ over source models, via Sherman–Morrison updates.
Ridge. `RidgeCV`, $\alpha \in \{10^{-2}, \ldots, 10^{3}\}$ (12 values), source coreset responses $\rightarrow$ source full accuracy.
Cross-fitting. 5 folds over coreset items.
AUTO. Inner splits use seed-dependent random permutations; 2 repetitions of coreset draws per inner split; inner IRT fits are refitted on the inner source subset.

C Additional results

Table 5: Relative RMSE at all five budgets (same format as Table 1); columns are protocol$\times n$ in the order $n = 10, 25, 50, 100, 200$.

	Random split					Held-out organisations					Stronger targets (extrapolation)				
	n=10	n=25	n=50	n=100	n=200	n=10	n=25	n=50	n=100	n=200	n=10	n=25	n=50	n=100	n=200
Random sampling	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Difficulty-stratified sampling	0.84 (0.79–0.88)	0.83 (0.79–0.88)	0.83 (0.79–0.87)	0.83 (0.79–0.87)	0.84 (0.80–0.88)	0.83 (0.76–0.91)	0.83 (0.76–0.89)	0.83 (0.77–0.89)	0.86 (0.80–0.91)	0.84 (0.79–0.89)	0.85 (0.80–0.92)	0.82 (0.78–0.87)	0.83 (0.78–0.88)	0.82 (0.78–0.87)	0.82 (0.78–0.86)
Random + linear calibration	0.56 (0.45–0.68)	0.67 (0.57–0.76)	0.73 (0.65–0.80)	0.79 (0.74–0.83)	0.81 (0.76–0.85)	0.48 (0.35–0.65)	0.61 (0.49–0.75)	0.68 (0.58–0.81)	0.77 (0.69–0.85)	0.79 (0.73–0.84)	0.85 (0.72–1.00)	0.95 (0.85–1.07)	1.01 (0.90–1.15)	1.07 (0.95–1.19)	1.06 (0.97–1.18)
Random + ridge regression	0.52 (0.41–0.64)	0.61 (0.50–0.72)	0.66 (0.55–0.78)	0.73 (0.63–0.83)	0.79 (0.68–0.91)	0.43 (0.32–0.60)	0.53 (0.41–0.67)	0.57 (0.46–0.73)	0.67 (0.56–0.81)	0.71 (0.62–0.85)	0.91 (0.73–1.11)	1.12 (0.94–1.32)	1.35 (1.13–1.61)	1.65 (1.37–1.96)	2.06 (1.68–2.52)
Anchor points (weighted mean)	0.68 (0.60–0.76)	0.78 (0.68–0.89)	0.91 (0.79–1.06)	1.08 (0.91–1.28)	1.28 (1.09–1.53)	0.67 (0.55–0.81)	0.82 (0.67–1.03)	0.94 (0.76–1.23)	1.21 (0.95–1.63)	1.41 (1.09–1.87)	0.72 (0.56–1.00)	0.90 (0.65–1.30)	1.08 (0.76–1.62)	1.34 (0.90–2.07)	1.67 (1.06–2.62)
Anchor points + IRT (tinyBenchmarks-style)	0.47 (0.37–0.59)	0.58 (0.47–0.71)	0.68 (0.54–0.85)	0.83 (0.66–1.07)	1.00 (0.78–1.33)	0.38 (0.31–0.48)	0.48 (0.40–0.58)	0.54 (0.46–0.65)	0.67 (0.58–0.81)	0.77 (0.66–0.88)	0.80 (0.63–0.99)	1.02 (0.85–1.22)	1.22 (1.03–1.43)	1.40 (1.11–1.73)	1.52 (1.08–2.05)
D-optimal items + IRT (info-based)	0.56 (0.43–0.73)	0.70 (0.55–0.92)	0.83 (0.64–1.10)	1.01 (0.79–1.33)	1.19 (0.92–1.59)	0.43 (0.32–0.55)	0.53 (0.41–0.67)	0.61 (0.48–0.76)	0.77 (0.63–0.96)	0.89 (0.73–1.09)	0.77 (0.62–0.96)	1.01 (0.85–1.20)	1.28 (1.07–1.53)	1.62 (1.33–1.98)	2.00 (1.58–2.55)
Random + IRT (pure)	0.56 (0.46–0.67)	0.65 (0.55–0.75)	0.69 (0.60–0.80)	0.78 (0.70–0.86)	0.91 (0.81–1.02)	0.48 (0.37–0.61)	0.57 (0.46–0.69)	0.61 (0.52–0.73)	0.71 (0.64–0.80)	0.81 (0.76–0.87)	0.92 (0.72–1.14)	1.08 (0.90–1.29)	1.25 (1.08–1.46)	1.54 (1.35–1.79)	2.05 (1.76–2.42)
Random + IRT-AIPW	0.73 (0.68–0.79)	0.78 (0.73–0.82)	0.78 (0.74–0.83)	0.80 (0.77–0.83)	0.78 (0.74–0.82)	0.70 (0.63–0.77)	0.77 (0.71–0.83)	0.78 (0.71–0.84)	0.80 (0.74–0.86)	0.78 (0.73–0.83)	0.74 (0.68–0.81)	0.78 (0.73–0.84)	0.79 (0.73–0.84)	0.82 (0.77–0.87)	0.83 (0.78–0.88)
VA-PPS + IRT-AIPW	0.74 (0.68–0.81)	0.78 (0.72–0.87)	0.80 (0.74–0.87)	0.82 (0.77–0.89)	0.80 (0.74–0.89)	0.71 (0.63–0.81)	0.75 (0.68–0.83)	0.74 (0.68–0.80)	0.77 (0.71–0.82)	0.76 (0.71–0.82)	0.72 (0.65–0.80)	0.76 (0.69–0.84)	0.80 (0.72–0.89)	0.88 (0.78–0.98)	0.84 (0.73–0.96)
VA-PPS + IRT (pure)	0.55 (0.45–0.67)	0.65 (0.54–0.79)	0.71 (0.60–0.85)	0.82 (0.69–1.00)	0.98 (0.81–1.23)	0.46 (0.36–0.58)	0.54 (0.45–0.64)	0.59 (0.51–0.68)	0.69 (0.64–0.76)	0.79 (0.74–0.85)	0.85 (0.68–1.07)	1.02 (0.86–1.20)	1.20 (1.04–1.40)	1.51 (1.30–1.76)	1.99 (1.71–2.35)
VA-PPS + Cal-IRT (ours)	0.61 (0.54–0.68)	0.69 (0.63–0.75)	0.71 (0.67–0.77)	0.75 (0.71–0.80)	0.74 (0.68–0.80)	0.56 (0.47–0.64)	0.65 (0.59–0.72)	0.69 (0.62–0.76)	0.74 (0.69–0.80)	0.73 (0.66–0.79)	0.68 (0.60–0.77)	0.70 (0.64–0.76)	0.72 (0.67–0.79)	0.76 (0.70–0.82)	0.74 (0.66–0.82)
VA-PPS + ShrinkCal-IRT (ours)	0.57 (0.50–0.66)	0.64 (0.58–0.72)	0.67 (0.61–0.74)	0.71 (0.66–0.77)	0.73 (0.67–0.80)	0.50 (0.41–0.59)	0.58 (0.51–0.65)	0.62 (0.54–0.69)	0.68 (0.62–0.74)	0.69 (0.64–0.75)	0.74 (0.62–0.87)	0.80 (0.71–0.90)	0.84 (0.76–0.92)	0.88 (0.82–0.93)	0.84 (0.78–0.91)
AUTO (pseudo-shift, minimax)	0.58 (0.48–0.69)	0.68 (0.59–0.76)	0.71 (0.64–0.79)	0.75 (0.71–0.80)	0.79 (0.71–0.90)	0.51 (0.38–0.66)	0.62 (0.51–0.73)	0.67 (0.58–0.75)	0.73 (0.67–0.78)	0.72 (0.66–0.79)	0.72 (0.61–0.84)	0.77 (0.68–0.87)	0.83 (0.74–0.91)	0.82 (0.74–0.91)	0.83 (0.74–0.95)
AUTO (pseudo-shift, mean)	0.55 (0.44–0.68)	0.66 (0.55–0.76)	0.70 (0.61–0.78)	0.72 (0.67–0.77)	0.78 (0.70–0.89)	0.47 (0.35–0.62)	0.58 (0.45–0.72)	0.62 (0.52–0.72)	0.70 (0.63–0.77)	0.71 (0.65–0.77)	0.75 (0.63–0.87)	0.81 (0.71–0.95)	0.85 (0.76–0.97)	0.87 (0.76–0.99)	0.83 (0.73–0.96)

Table 6: Spearman correlation between predicted and true accuracy of the target models in a split (mean over benchmarks and seeds; in parentheses the difference to random sampling with its 95% hierarchical-bootstrap interval).

	Random split		Held-out organisations		Stronger targets (extrapolation)	
	n=25	n=100	n=25	n=100	n=25	n=100
Random sampling	0.795	0.910	0.726	0.871	0.532	0.759
Difficulty-stratified sampling	0.784 (-0.011; -0.036..+0.006)	0.913 (+0.003; -0.008..+0.015)	0.737 (+0.011; -0.010..+0.032)	0.873 (+0.002; -0.010..+0.016)	0.536 (+0.004; -0.030..+0.038)	0.774 (+0.015; -0.007..+0.039)
Random + linear calibration	0.795 (+0.000; +0.000..+0.000)	0.910 (+0.000; +0.000..+0.000)	0.726 (+0.000; +0.000..+0.000)	0.871 (+0.000; +0.000..+0.000)	0.529 (-0.003; -0.008..+0.000)	0.759 (+0.000; +0.000..+0.000)
Random + ridge regression	0.854 (+0.059; +0.019..+0.108)	0.941 (+0.031; +0.001..+0.073)	0.817 (+0.091; +0.022..+0.173)	0.922 (+0.051; +0.004..+0.113)	0.404 (-0.128; -0.202..-0.064)	0.624 (-0.135; -0.200..-0.064)
Anchor points (weighted mean)	0.844 (+0.048; +0.004..+0.078)	0.899 (-0.011; -0.054..+0.017)	0.776 (+0.049; -0.011..+0.090)	0.835 (-0.036; -0.118..+0.020)	0.481 (-0.050; -0.098..-0.002)	0.701 (-0.058; -0.120..+0.007)
Anchor points + IRT (tinyBenchmarks-style)	0.892 (+0.097; +0.056..+0.143)	0.947 (+0.037; +0.014..+0.069)	0.854 (+0.128; +0.075..+0.189)	0.921 (+0.051; +0.019..+0.091)	0.403 (-0.128; -0.216..-0.049)	0.680 (-0.079; -0.219..+0.055)
D-optimal items + IRT (info-based)	0.879 (+0.084; +0.040..+0.132)	0.938 (+0.029; +0.000..+0.067)	0.856 (+0.130; +0.063..+0.201)	0.918 (+0.047; +0.005..+0.099)	0.415 (-0.117; -0.262..+0.009)	0.603 (-0.156; -0.334..-0.012)
Random + IRT (pure)	0.841 (+0.046; +0.020..+0.077)	0.930 (+0.020; +0.004..+0.042)	0.788 (+0.062; +0.021..+0.107)	0.898 (+0.027; +0.005..+0.055)	0.407 (-0.125; -0.191..-0.059)	0.614 (-0.146; -0.248..-0.051)
Random + IRT-AIPW	0.803 (+0.007; +0.000..+0.015)	0.917 (+0.008; +0.001..+0.018)	0.733 (+0.007; -0.001..+0.018)	0.882 (+0.011; +0.001..+0.026)	0.515 (-0.017; -0.027..-0.005)	0.760 (+0.000; -0.009..+0.013)
VA-PPS + IRT-AIPW	0.820 (+0.024; +0.009..+0.039)	0.925 (+0.016; +0.004..+0.032)	0.760 (+0.034; +0.019..+0.051)	0.896 (+0.025; +0.008..+0.047)	0.551 (+0.020; -0.009..+0.054)	0.777 (+0.017; -0.007..+0.048)
VA-PPS + IRT (pure)	0.854 (+0.059; +0.031..+0.090)	0.934 (+0.024; +0.006..+0.050)	0.808 (+0.082; +0.044..+0.124)	0.903 (+0.032; +0.011..+0.062)	0.436 (-0.096; -0.169..-0.031)	0.642 (-0.117; -0.214..-0.023)
VA-PPS + Cal-IRT (ours)	0.832 (+0.036; +0.020..+0.054)	0.930 (+0.020; +0.007..+0.039)	0.776 (+0.050; +0.029..+0.074)	0.899 (+0.029; +0.009..+0.054)	0.555 (+0.024; -0.004..+0.058)	0.787 (+0.028; +0.004..+0.059)
VA-PPS + ShrinkCal-IRT (ours)	0.848 (+0.052; +0.030..+0.075)	0.937 (+0.027; +0.009..+0.052)	0.800 (+0.074; +0.042..+0.113)	0.911 (+0.040; +0.013..+0.073)	0.546 (+0.014; -0.015..+0.052)	0.784 (+0.024; +0.001..+0.056)
AUTO (pseudo-shift, minimax)	0.831 (+0.036; +0.016..+0.058)	0.929 (+0.019; +0.004..+0.040)	0.787 (+0.060; +0.028..+0.101)	0.903 (+0.032; +0.007..+0.063)	0.528 (-0.004; -0.037..+0.034)	0.774 (+0.015; -0.023..+0.053)

Table 7: Kendall τ, same format as Table 6.

	Random split		Held-out organisations		Stronger targets (extrapolation)	
	n=25	n=100	n=25	n=100	n=25	n=100
Random sampling	0.660	0.798	0.588	0.742	0.413	0.612
Difficulty-stratified sampling	0.653 (-0.007; -0.027..+0.008)	0.801 (+0.004; -0.007..+0.015)	0.600 (+0.012; -0.006..+0.031)	0.744 (+0.002; -0.010..+0.015)	0.416 (+0.003; -0.026..+0.032)	0.625 (+0.013; -0.004..+0.035)
Random + linear calibration	0.660 (+0.000; +0.000..+0.000)	0.798 (+0.000; +0.000..+0.000)	0.588 (+0.000; +0.000..+0.000)	0.742 (+0.000; +0.000..+0.000)	0.411 (-0.002; -0.006..+0.000)	0.612 (+0.000; +0.000..+0.000)
Random + ridge regression	0.696 (+0.035; +0.004..+0.075)	0.822 (+0.024; -0.006..+0.064)	0.649 (+0.061; +0.008..+0.124)	0.786 (+0.045; -0.001..+0.099)	0.293 (-0.120; -0.180..-0.070)	0.471 (-0.141; -0.198..-0.081)
Anchor points (weighted mean)	0.707 (+0.047; +0.006..+0.075)	0.794 (-0.004; -0.045..+0.027)	0.633 (+0.045; -0.007..+0.083)	0.716 (-0.026; -0.100..+0.028)	0.351 (-0.063; -0.103..-0.024)	0.542 (-0.069; -0.127..-0.008)
Anchor points + IRT (tinyBenchmarks-style)	0.747 (+0.087; +0.054..+0.120)	0.840 (+0.042; +0.020..+0.069)	0.698 (+0.111; +0.075..+0.148)	0.797 (+0.056; +0.030..+0.083)	0.290 (-0.123; -0.188..-0.063)	0.527 (-0.084; -0.189..+0.027)
D-optimal items + IRT (info-based)	0.725 (+0.065; +0.030..+0.103)	0.823 (+0.025; -0.003..+0.058)	0.694 (+0.106; +0.055..+0.155)	0.784 (+0.043; +0.004..+0.085)	0.310 (-0.103; -0.205..-0.006)	0.473 (-0.139; -0.270..-0.025)
Random + IRT (pure)	0.685 (+0.025; +0.004..+0.049)	0.812 (+0.015; -0.000..+0.033)	0.624 (+0.037; +0.005..+0.067)	0.762 (+0.020; +0.000..+0.043)	0.296 (-0.117; -0.170..-0.064)	0.468 (-0.144; -0.218..-0.068)
Random + IRT-AIPW	0.647 (-0.013; -0.019..-0.005)	0.797 (-0.001; -0.008..+0.009)	0.573 (-0.014; -0.021..-0.004)	0.742 (+0.001; -0.009..+0.012)	0.379 (-0.034; -0.043..-0.024)	0.596 (-0.015; -0.026..-0.003)
VA-PPS + IRT-AIPW	0.668 (+0.007; -0.007..+0.022)	0.810 (+0.012; +0.001..+0.025)	0.602 (+0.014; +0.001..+0.026)	0.761 (+0.019; +0.005..+0.035)	0.408 (-0.005; -0.030..+0.021)	0.609 (-0.003; -0.028..+0.024)
VA-PPS + IRT (pure)	0.699 (+0.039; +0.015..+0.063)	0.818 (+0.021; +0.003..+0.040)	0.646 (+0.059; +0.032..+0.087)	0.771 (+0.029; +0.011..+0.051)	0.317 (-0.096; -0.152..-0.048)	0.493 (-0.119; -0.195..-0.047)
VA-PPS + Cal-IRT (ours)	0.679 (+0.019; +0.004..+0.034)	0.815 (+0.017; +0.006..+0.031)	0.615 (+0.028; +0.012..+0.044)	0.764 (+0.023; +0.007..+0.042)	0.411 (-0.003; -0.027..+0.025)	0.621 (+0.009; -0.015..+0.036)
VA-PPS + ShrinkCal-IRT (ours)	0.696 (+0.036; +0.016..+0.054)	0.824 (+0.026; +0.009..+0.046)	0.640 (+0.052; +0.029..+0.076)	0.780 (+0.038; +0.017..+0.063)	0.402 (-0.011; -0.035..+0.018)	0.617 (+0.005; -0.018..+0.033)
AUTO (pseudo-shift, minimax)	0.676 (+0.016; -0.001..+0.034)	0.812 (+0.014; +0.001..+0.030)	0.623 (+0.036; +0.011..+0.065)	0.767 (+0.025; +0.004..+0.050)	0.387 (-0.026; -0.056..+0.006)	0.611 (-0.001; -0.030..+0.032)

D Shift severity

Targets are the strongest 10% / 25% / 50% or the weakest 25% of models (by full-benchmark accuracy); the remaining models are sources. Ten seeds (0–9), $n \in \{25, 100\}$, designs random, stratified, VA-PPS ($\lambda = 1$) and anchors; the "Stronger targets" (25%) column is the main-run result restricted to the same seeds.

E Number of source models

Seven benchmarks (ARC-Challenge, GPQA, IFEval, MuSR, MATH Lvl 5, HELM MMLU, HELM GSM), seeds 0–9; m source models drawn at random from the split's source set ("all" = no subsampling; HELM splits have fewer than 100 sources, so $m = 100$ equals "all" there). Relative RMSE vs. random sampling on the same targets (random sampling does not depend on m).

Table 8: Paired comparisons: ratio of relative RMSEs (first method / second method) with 95% hierarchical-bootstrap intervals; * = interval excludes 1.

	Random split		Held-out organisations		Stronger targets (extrapolation)	
	n=25	n=100	n=25	n=100	n=25	n=100
VA-PPS + Cal-IRT (ours) vs. Random + IRT-AIPW	0.89 (0.85–0.93)*	0.94 (0.90–0.98)*	0.85 (0.81–0.88)*	0.93 (0.88–0.97)*	0.89 (0.84–0.95)*	0.92 (0.87–0.97)*
VA-PPS + Cal-IRT (ours) vs. Difficulty-stratified sampling	0.83 (0.77–0.89)*	0.91 (0.86–0.95)*	0.79 (0.73–0.85)*	0.87 (0.83–0.91)*	0.85 (0.80–0.91)*	0.92 (0.87–0.98)*
VA-PPS + Cal-IRT (ours) vs. Random + ridge regression	1.14 (1.00–1.34)*	1.04 (0.92–1.19)	1.24 (1.00–1.56)*	1.12 (0.92–1.34)	0.62 (0.55–0.72)*	0.46 (0.40–0.53)*
VA-PPS + ShrinkCal-IRT (ours) vs. Random + ridge regression	1.06 (0.97–1.21)	0.98 (0.89–1.10)	1.11 (0.93–1.34)	1.02 (0.88–1.19)	0.72 (0.66–0.78)*	0.53 (0.47–0.61)*
VA-PPS + ShrinkCal-IRT (ours) vs. VA-PPS + Cal-IRT (ours)	0.93 (0.89–0.97)*	0.95 (0.91–0.99)*	0.89 (0.85–0.94)*	0.91 (0.88–0.96)*	1.15 (1.07–1.22)*	1.15 (1.11–1.22)*
VA-PPS + Cal-IRT (ours) vs. VA-PPS + IRT-AIPW	0.88 (0.83–0.93)*	0.91 (0.87–0.96)*	0.87 (0.80–0.93)*	0.96 (0.94–0.99)*	0.92 (0.86–0.98)*	0.87 (0.79–0.94)*
VA-PPS + IRT-AIPW vs. Difficulty-stratified sampling	0.94 (0.87–1.03)	0.99 (0.92–1.08)	0.91 (0.86–0.98)*	0.90 (0.87–0.93)*	0.93 (0.86–1.01)	1.06 (0.96–1.21)
AUTO (pseudo-shift, minimax) vs. VA-PPS + Cal-IRT (ours)	0.98 (0.91–1.05)	1.00 (0.96–1.05)	0.95 (0.84–1.07)	0.98 (0.94–1.03)	1.11 (1.04–1.18)*	1.08 (1.02–1.15)*

Table 9: Design effects with the estimator held fixed (ratio of RMSEs, design A / design B, paired bootstrap). Names `design/estimator`; digit = IRT dimension.

	Random split		Held-out organisations		Stronger targets (extrapolation)	
	n=25	n=100	n=25	n=100	n=25	n=100
strat/cal4 vs. rand/cal4	1.00 (0.98–1.03)	0.99 (0.96–1.01)	0.98 (0.94–1.01)	1.00 (0.97–1.04)	0.98 (0.95–1.02)	0.96 (0.93–1.00)*
pps0.5/cal4 vs. strat/cal4	1.00 (0.97–1.03)	0.98 (0.94–1.01)	0.99 (0.96–1.03)	0.98 (0.95–1.00)	0.95 (0.92–1.00)	0.99 (0.95–1.03)
VA-PPS + Cal-IRT (ours) vs. strat/cal4	0.98 (0.93–1.02)	0.98 (0.93–1.03)	0.96 (0.91–0.99)*	0.95 (0.90–0.99)*	0.93 (0.87–0.98)*	0.99 (0.94–1.04)
VA-PPS + Cal-IRT (ours) vs. rand/cal4	0.98 (0.94–1.02)	0.97 (0.93–1.01)	0.94 (0.90–0.97)*	0.95 (0.90–0.99)*	0.91 (0.86–0.97)*	0.95 (0.90–1.00)
strat/shrcal4 vs. rand/shrcal4	1.00 (0.97–1.02)	0.99 (0.96–1.01)	0.98 (0.94–1.02)	1.00 (0.98–1.03)	0.99 (0.95–1.03)	0.96 (0.93–1.00)*
pps0.5/shrcal4 vs. strat/shrcal4	0.99 (0.95–1.02)	0.98 (0.95–1.01)	0.98 (0.95–1.02)	0.96 (0.94–0.99)*	0.97 (0.93–1.01)	1.00 (0.97–1.03)
VA-PPS + ShrinkCal-IRT (ours) vs. strat/shrcal4	0.97 (0.93–1.02)	0.98 (0.93–1.03)	0.95 (0.90–0.98)*	0.95 (0.91–0.98)*	0.96 (0.92–1.01)	1.03 (0.99–1.06)
VA-PPS + ShrinkCal-IRT (ours) vs. rand/shrcal4	0.97 (0.93–1.01)	0.97 (0.93–1.01)	0.93 (0.90–0.97)*	0.95 (0.90–0.99)*	0.95 (0.91–1.00)*	0.99 (0.95–1.03)
strat/greg4 vs. Random + IRT-AIPW	1.00 (0.97–1.03)	0.99 (0.96–1.01)	0.98 (0.94–1.02)	1.00 (0.98–1.04)	0.99 (0.95–1.03)	0.96 (0.92–1.00)*
pps0.5/greg4 vs. strat/greg4	0.98 (0.95–1.01)	0.96 (0.93–1.00)*	0.98 (0.96–1.02)	0.97 (0.94–1.00)*	0.95 (0.92–0.99)*	0.99 (0.94–1.03)
VA-PPS + IRT-AIPW vs. strat/greg4	1.01 (0.94–1.10)	1.04 (0.97–1.12)	1.00 (0.96–1.05)	0.96 (0.93–0.99)*	0.98 (0.91–1.04)	1.11 (1.00–1.25)
VA-PPS + IRT-AIPW vs. Random + IRT-AIPW	1.01 (0.95–1.08)	1.03 (0.97–1.10)	0.98 (0.93–1.05)	0.96 (0.93–0.99)*	0.97 (0.88–1.06)	1.07 (0.96–1.19)
Difficulty-stratified sampling vs. Random sampling	0.83 (0.79–0.88)*	0.83 (0.79–0.87)*	0.83 (0.76–0.89)*	0.86 (0.80–0.91)*	0.82 (0.78–0.87)*	0.82 (0.78–0.87)*
pps0.5/mean vs. Difficulty-stratified sampling	1.00 (0.97–1.03)	0.98 (0.94–1.02)	1.01 (0.97–1.05)	0.97 (0.94–1.00)*	0.99 (0.96–1.03)	1.01 (0.97–1.05)
pps1.0/mean vs. Difficulty-stratified sampling	1.14 (1.06–1.25)*	1.28 (1.11–1.45)*	1.14 (1.04–1.25)*	1.09 (1.00–1.23)	1.18 (1.06–1.30)*	1.47 (1.20–1.82)*
pps1.0/mean vs. Random sampling	0.95 (0.89–1.02)	1.06 (0.94–1.20)	0.94 (0.87–1.01)	0.94 (0.87–1.03)	0.97 (0.88–1.05)	1.21 (1.00–1.51)*

F Reproducibility and disclosures

Code and commands. See `README.md` in the supplementary code. In short: `run_exp.py -tag main -seeds 0-29 -budgets 10,25,50,100,200 -auto -jobs 4` (main grid, 990 jobs, about 3.5 h on 4 CPU cores), `run_exp.py -tag shift -protocols top10,top50,bottom -seeds 0-9 -budgets 25,100 -designs rand,strat,pps1.0,anchor` (shift-severity run), and the source-size runs (`run_exp.py -tag src`K `-benchmarks arc_challenge,gpqa,ifeval,musr,openllm_math,mmlu,gsm -protocols random,top -seeds 0-9 -budgets 25,100 -designs strat,pps1.0 -max_sources` K, $K \in \{20, 50, 100\}$); then `analyze.py main`, `analyze.py shift` and `analyze_src.py` regenerate all tables and figures. Re-running two main-grid jobs (IFEval and GSM, stronger-target split, seed 3) reproduced all stored results exactly (maximum absolute difference 0). Software: Python 3.11, numpy 2.4.6, scipy 1.17.1, scikit-learn 1.9.1, pandas 3.1.0, joblib 1.6.0, matplotlib 3.11.2. Randomness: split seed s uses `default_rng(10000+s)`; coreset/IRT-initialisation randomness uses `default_rng(s)`; AUTO inner splits use `default_rng(777000+s)`. Development runs used seeds ≥ 1000 (and 2000–2007) and are not part of the reported tables.

Data. Files `data/scores/*.jbl` from `https://github.com/socialfoundations/benchmark-prediction` at commit `55faff496947789f94e6b5908c77cbc3e84f81ad` (SHA-256 sums of all files are recorded in `data/SHA256SUMS.txt`). We only used the score matrices and model lists; we did not use the `benchpred` package, nor re-run any model. Upstream sources: Open LLM Leaderboard v2, HELM.

Related-work disclosure. Regarding the restriction on reading the paper that studies this exact problem: we did not search for, open or read any paper on that topic apart from the items of the provided starter list. The starter list includes the NeurIPS 2025 paper of Zhang et al. [2025]; we did *not* read it. Our knowledge of it is limited to its description in the starter list and the README of the data repository (which we read when downloading the data and which summarises its conclusions in three sentences, quoted in spirit in Section 2). Other literature was located through a scholarly-search tool (titles and abstract snippets only) to verify citation details. No other encounter with restricted content occurred.

Use of an AI agent. All code, experiments, analysis and text of this study were produced by an autonomous AI research agent (Claude Sonnet 5.5, Claude Agent SDK) running unattended.

Failed or abandoned items. The first launch of the main run used 38 worker processes under a 4-core CPU quota and was killed and restarted with 4 workers before producing any result file. The

Table 10: Number of benchmarks on which the method has lower seed-averaged MSE than random sampling.

	Random split		Held-out organisations		Stronger targets (extrapolation)	
	n=25	n=100	n=25	n=100	n=25	n=100
Difficulty-stratified sampling	13/13	13/13	7/7	7/7	13/13	13/13
Random + linear calibration	13/13	13/13	7/7	7/7	9/13	5/13
Random + ridge regression	13/13	13/13	7/7	7/7	4/13	2/13
Anchor points (weighted mean)	10/13	6/13	5/7	2/7	9/13	6/13
Anchor points + IRT (tinyBenchmarks-style)	12/13	9/13	7/7	6/7	5/13	3/13
D-optimal items + IRT (info-based)	11/13	7/13	7/7	6/7	6/13	2/13
Random + IRT (pure)	12/13	12/13	7/7	7/7	4/13	1/13
Random + IRT-AIPW	13/13	13/13	7/7	7/7	13/13	13/13
VA-PPS + IRT-AIPW	12/13	11/13	7/7	7/7	12/13	10/13
VA-PPS + IRT (pure)	12/13	10/13	7/7	7/7	5/13	2/13
VA-PPS + Cal-IRT (ours)	13/13	13/13	7/7	7/7	13/13	13/13
VA-PPS + ShrinkCal-IRT (ours)	13/13	13/13	7/7	7/7	13/13	13/13
AUTO (pseudo-shift, minimax)	13/13	13/13	7/7	7/7	12/13	11/13
AUTO (pseudo-shift, mean)	13/13	13/13	7/7	7/7	11/13	10/13

Table 11: Ablation: design $\times$ estimator at $d = 4$ (relative RMSE; `design/estimator`; `rand`, `strat`, `pps`λ = VA-PPS with floor parameter λ; estimators: mean, pure IRT, plug-in, AIPW (`greg`), cross-fitted AIPW, Cal-IRT, Cal-IRT+AIPW, ShrinkCal). Rows named with prose are the main-table entries.

	Random split		Held-out organisations		Stronger targets (extrapolation)	
	n=25	n=100	n=25	n=100	n=25	n=100
Random sampling	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Random + IRT (pure)	0.65 (0.55–0.75)	0.78 (0.70–0.86)	0.57 (0.46–0.69)	0.71 (0.64–0.80)	1.08 (0.90–1.29)	1.54 (1.35–1.79)
rand/plug4	0.64 (0.55–0.75)	0.75 (0.68–0.82)	0.57 (0.46–0.68)	0.69 (0.62–0.78)	1.06 (0.89–1.26)	1.41 (1.23–1.65)
Random + IRT-AIPW	0.78 (0.73–0.82)	0.80 (0.77–0.83)	0.77 (0.71–0.83)	0.80 (0.74–0.86)	0.78 (0.73–0.84)	0.82 (0.77–0.87)
rand/gregcf4	0.87 (0.82–0.92)	0.83 (0.80–0.86)	0.86 (0.81–0.92)	0.83 (0.76–0.88)	0.87 (0.81–0.93)	0.86 (0.80–0.92)
rand/cal4	0.71 (0.65–0.77)	0.78 (0.74–0.81)	0.70 (0.62–0.78)	0.78 (0.72–0.84)	0.77 (0.70–0.83)	0.80 (0.75–0.84)
rand/calgreg4	0.77 (0.73–0.82)	0.80 (0.77–0.83)	0.77 (0.71–0.84)	0.81 (0.74–0.86)	0.78 (0.73–0.83)	0.81 (0.76–0.86)
rand/shrcal4	0.66 (0.60–0.73)	0.73 (0.69–0.78)	0.63 (0.54–0.71)	0.72 (0.65–0.78)	0.84 (0.75–0.94)	0.89 (0.83–0.94)
Difficulty-stratified sampling	0.83 (0.79–0.88)	0.83 (0.79–0.87)	0.83 (0.76–0.89)	0.86 (0.80–0.91)	0.82 (0.78–0.87)	0.82 (0.78–0.87)
strat/irt4	0.65 (0.55–0.75)	0.78 (0.70–0.86)	0.56 (0.46–0.68)	0.72 (0.65–0.80)	1.07 (0.89–1.27)	1.54 (1.33–1.79)
strat/plug4	0.64 (0.55–0.74)	0.75 (0.68–0.82)	0.56 (0.45–0.68)	0.70 (0.62–0.79)	1.05 (0.87–1.25)	1.39 (1.21–1.64)
strat/greg4	0.78 (0.73–0.83)	0.79 (0.75–0.82)	0.75 (0.69–0.83)	0.80 (0.75–0.85)	0.78 (0.73–0.84)	0.79 (0.74–0.84)
strat/gregcf4	0.87 (0.81–0.92)	0.82 (0.79–0.86)	0.85 (0.78–0.93)	0.83 (0.78–0.88)	0.89 (0.83–0.95)	0.84 (0.78–0.89)
strat/cal4	0.71 (0.65–0.77)	0.77 (0.73–0.81)	0.68 (0.60–0.76)	0.78 (0.73–0.83)	0.75 (0.68–0.83)	0.76 (0.71–0.82)
strat/calgreg4	0.78 (0.73–0.83)	0.79 (0.75–0.82)	0.76 (0.69–0.83)	0.80 (0.75–0.85)	0.78 (0.73–0.83)	0.78 (0.73–0.83)
strat/shrcal4	0.66 (0.60–0.73)	0.73 (0.68–0.78)	0.61 (0.53–0.70)	0.72 (0.66–0.78)	0.83 (0.73–0.93)	0.85 (0.80–0.91)
pps0.5/mean	0.83 (0.79–0.88)	0.81 (0.77–0.86)	0.83 (0.77–0.89)	0.83 (0.78–0.88)	0.82 (0.77–0.87)	0.83 (0.78–0.88)
pps0.5/irt4	0.65 (0.54–0.77)	0.79 (0.69–0.92)	0.56 (0.46–0.66)	0.70 (0.64–0.78)	1.03 (0.86–1.23)	1.53 (1.33–1.77)
pps0.5/plug4	0.65 (0.54–0.77)	0.76 (0.67–0.88)	0.56 (0.45–0.66)	0.68 (0.61–0.76)	1.01 (0.84–1.20)	1.38 (1.19–1.61)
pps0.5/greg4	0.76 (0.71–0.82)	0.76 (0.72–0.80)	0.74 (0.67–0.81)	0.78 (0.73–0.82)	0.74 (0.69–0.80)	0.78 (0.73–0.83)
pps0.5/gregcf4	0.84 (0.79–0.90)	0.79 (0.75–0.83)	0.83 (0.76–0.90)	0.81 (0.76–0.85)	0.83 (0.79–0.89)	0.82 (0.77–0.88)
pps0.5/cal4	0.70 (0.64–0.77)	0.75 (0.71–0.79)	0.68 (0.60–0.75)	0.76 (0.71–0.81)	0.72 (0.66–0.79)	0.76 (0.71–0.81)
pps0.5/calgreg4	0.76 (0.71–0.82)	0.76 (0.72–0.80)	0.75 (0.68–0.81)	0.78 (0.73–0.82)	0.74 (0.69–0.80)	0.77 (0.72–0.83)
pps0.5/shrcal4	0.65 (0.58–0.73)	0.71 (0.66–0.76)	0.60 (0.52–0.68)	0.69 (0.64–0.75)	0.80 (0.71–0.91)	0.85 (0.80–0.92)
pps1.0/mean	0.95 (0.89–1.02)	1.06 (0.94–1.20)	0.94 (0.87–1.01)	0.94 (0.87–1.03)	0.97 (0.88–1.05)	1.21 (1.00–1.51)
VA-PPS + IRT (pure)	0.65 (0.54–0.79)	0.82 (0.69–1.00)	0.54 (0.45–0.64)	0.69 (0.64–0.76)	1.02 (0.86–1.20)	1.51 (1.30–1.76)
pps1.0/plug4	0.65 (0.54–0.78)	0.79 (0.67–0.95)	0.54 (0.45–0.64)	0.67 (0.61–0.74)	0.99 (0.84–1.18)	1.35 (1.16–1.59)
VA-PPS + IRT-AIPW	0.78 (0.72–0.87)	0.82 (0.77–0.89)	0.75 (0.68–0.83)	0.77 (0.71–0.82)	0.76 (0.69–0.84)	0.88 (0.78–0.98)
VA-PPS + Cal-IRT (ours)	0.69 (0.63–0.75)	0.75 (0.71–0.80)	0.65 (0.59–0.72)	0.74 (0.69–0.80)	0.70 (0.64–0.76)	0.76 (0.70–0.82)
pps1.0/calgreg4	0.77 (0.71–0.84)	0.81 (0.76–0.88)	0.75 (0.68–0.83)	0.77 (0.71–0.82)	0.76 (0.68–0.83)	0.87 (0.78–0.97)
VA-PPS + ShrinkCal-IRT (ours)	0.64 (0.58–0.72)	0.71 (0.66–0.77)	0.58 (0.51–0.65)	0.68 (0.62–0.74)	0.80 (0.71–0.90)	0.88 (0.82–0.93)

cross-fitted AIPW variant (`gregcf`) was worse than in-sample AIPW and is reported in Table 11 only. On the development run, weaker item regularisation (0.2 instead of 1) was slightly worse for Cal-IRT and larger ability-prior variances (25, 1000) were similar or worse, so neither was adopted; extra Platt penalties (5, 20) and an intercept-only recalibration were not better than the default either.

Table 12: Ablation: IRT dimension $d \in \{2, 4, 8\}$ for VA-PPS ($\lambda = 1$) with pure, Cal, ShrinkCal and AIPW estimators.

	Random split		Held-out organisations		Stronger targets (extrapolation)	
	n=25	n=100	n=25	n=100	n=25	n=100
Random sampling	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)
pps1.0/irt2	0.65 (0.54–0.78)	0.83 (0.71–1.02)	0.54 (0.44–0.64)	0.71 (0.65–0.79)	1.10 (0.93–1.30)	1.77 (1.49–2.10)
VA-PPS + IRT (pure)	0.65 (0.54–0.79)	0.82 (0.69–1.00)	0.54 (0.45–0.64)	0.69 (0.64–0.76)	1.02 (0.86–1.20)	1.51 (1.30–1.76)
pps1.0/irt8	0.68 (0.57–0.82)	0.83 (0.69–1.02)	0.58 (0.48–0.67)	0.69 (0.61–0.78)	1.02 (0.84–1.23)	1.39 (1.17–1.65)
pps1.0/cal2	0.70 (0.64–0.76)	0.76 (0.72–0.80)	0.67 (0.60–0.74)	0.76 (0.70–0.82)	0.69 (0.63–0.75)	0.76 (0.71–0.82)
VA-PPS + Cal-IRT (ours)	0.69 (0.63–0.75)	0.75 (0.71–0.80)	0.65 (0.59–0.72)	0.74 (0.69–0.80)	0.70 (0.64–0.76)	0.76 (0.70–0.82)
pps1.0/cal8	0.69 (0.62–0.77)	0.74 (0.70–0.80)	0.66 (0.57–0.74)	0.73 (0.67–0.79)	0.77 (0.70–0.86)	0.78 (0.71–0.85)
pps1.0/shrcal2	0.64 (0.58–0.71)	0.72 (0.67–0.77)	0.59 (0.51–0.65)	0.69 (0.63–0.75)	0.80 (0.72–0.88)	0.88 (0.82–0.93)
VA-PPS + ShrinkCal-IRT (ours)	0.64 (0.58–0.72)	0.71 (0.66–0.77)	0.58 (0.51–0.65)	0.68 (0.62–0.74)	0.80 (0.71–0.90)	0.88 (0.82–0.93)
pps1.0/shrcal8	0.66 (0.58–0.75)	0.72 (0.66–0.78)	0.60 (0.51–0.69)	0.68 (0.61–0.74)	0.85 (0.74–0.97)	0.88 (0.81–0.96)
pps1.0/greg2	0.80 (0.74–0.89)	0.83 (0.78–0.90)	0.77 (0.70–0.86)	0.78 (0.72–0.84)	0.78 (0.70–0.85)	0.89 (0.79–0.99)
VA-PPS + IRT-AIPW	0.78 (0.72–0.87)	0.82 (0.77–0.89)	0.75 (0.68–0.83)	0.77 (0.71–0.82)	0.76 (0.69–0.84)	0.88 (0.78–0.98)
pps1.0/greg8	0.77 (0.70–0.85)	0.82 (0.76–0.89)	0.74 (0.66–0.83)	0.76 (0.70–0.81)	0.78 (0.71–0.86)	0.88 (0.78–0.99)

Table 13: Other combinations: anchors and D-optimal items with different estimators, ridge/linear with other designs.

	Random split		Held-out organisations		Stronger targets (extrapolation)	
	n=25	n=100	n=25	n=100	n=25	n=100
Random sampling	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)
anchor/cal4	0.64 (0.58–0.71)	0.77 (0.71–0.84)	0.64 (0.55–0.73)	0.77 (0.67–0.89)	0.77 (0.66–0.92)	0.80 (0.67–0.95)
anchor/ridge	0.51 (0.41–0.63)	0.73 (0.58–0.90)	0.42 (0.33–0.54)	0.61 (0.48–0.80)	1.06 (0.86–1.32)	1.61 (1.29–2.03)
anchor/greg4	0.69 (0.61–0.79)	0.94 (0.81–1.10)	0.72 (0.60–0.87)	1.01 (0.82–1.31)	0.96 (0.75–1.35)	1.26 (0.92–1.87)
dopt4/cal	0.76 (0.65–0.90)	0.91 (0.80–1.06)	0.68 (0.54–0.81)	0.90 (0.71–1.10)	0.85 (0.69–1.05)	1.07 (0.82–1.39)
dopt4/irt	0.71 (0.55–0.93)	1.03 (0.81–1.35)	0.53 (0.41–0.67)	0.79 (0.65–0.97)	1.02 (0.86–1.22)	1.75 (1.48–2.09)
dopt2/cal	0.79 (0.67–0.93)	0.99 (0.86–1.13)	0.73 (0.56–0.90)	1.04 (0.78–1.33)	0.83 (0.69–1.00)	1.32 (1.06–1.67)
dopt8/cal	0.85 (0.69–1.05)	0.90 (0.78–1.05)	0.71 (0.56–0.88)	0.84 (0.71–0.99)	0.86 (0.69–1.07)	0.99 (0.81–1.23)
strat/ridge	0.60 (0.50–0.72)	0.72 (0.62–0.82)	0.51 (0.40–0.66)	0.66 (0.56–0.81)	1.11 (0.93–1.32)	1.62 (1.35–1.94)
pps1.0/ridge	0.61 (0.49–0.73)	0.73 (0.62–0.86)	0.50 (0.40–0.63)	0.64 (0.54–0.76)	1.05 (0.89–1.25)	1.54 (1.28–1.87)
pps1.0/lin	0.67 (0.57–0.78)	0.79 (0.73–0.86)	0.60 (0.47–0.74)	0.74 (0.67–0.81)	0.89 (0.79–1.00)	1.02 (0.89–1.14)

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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   Justification: The abstract and introduction state the scope (13 binary item-level benchmarks, three split types), the components proposed, and the negative findings (uncorrected predictors fail under extrapolation, pseudo-shift tuning does not help); Section 5 supports each claim.

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   Justification: Section 6 lists limitations (scope, shift operationalisation, re-implemented baselines, tuning, lack of guarantees).

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Table 14: Share (%) of AUTO (minimax) selections by estimator family (design and d marginalised).

split	n	Cal	ShrinkCal	Cal+AIPW	AIPW	plug-in	IRT	ridge	linear	mean
Held-out organisations	25	44	22	4	5	18	5	0	0	1
Held-out organisations	100	47	28	7	9	3	0	0	2	5
Random split	25	44	17	9	11	9	2	0	2	5
Random split	100	48	18	10	11	2	0	0	2	9
Stronger targets (extrapolation)	25	43	16	2	3	19	1	11	3	2
Stronger targets (extrapolation)	100	45	18	7	4	8	0	9	3	5

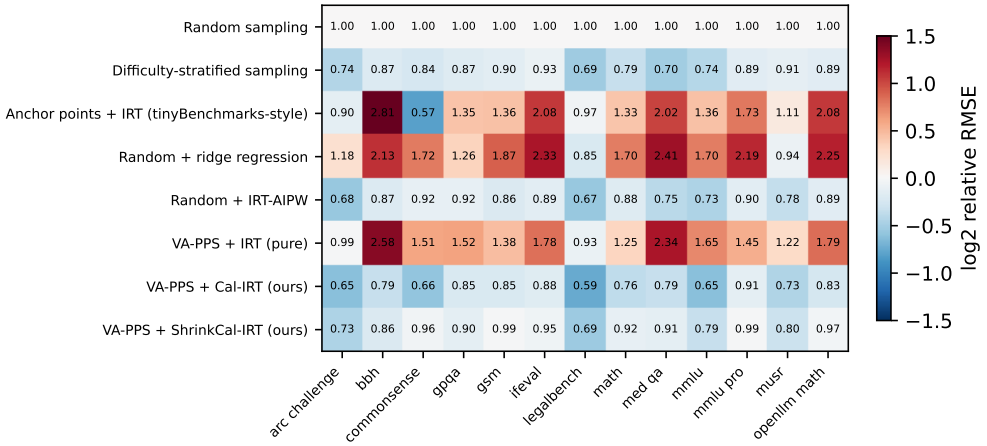


Figure 2: Per-benchmark relative RMSE at $n = 100$, stronger-target split (cell values; colour = $\log_2$).

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   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

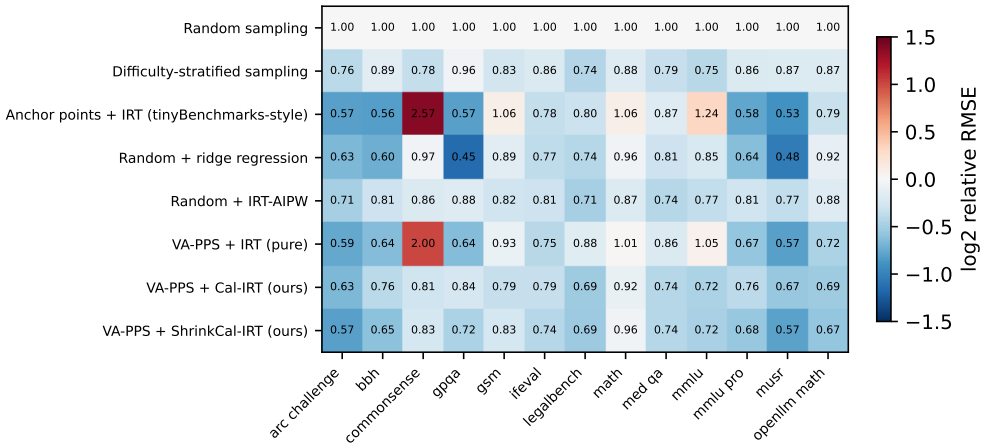


Figure 3: Per-benchmark relative RMSE at $n = 100$, random split.

Table 15: Relative RMSE vs. random sampling under different shift severities (geometric mean over 13 benchmarks; 95% hierarchical-bootstrap intervals).

	Top-10% targets		Stronger targets (extrapolation)		Top-50% targets		Weakest-25% targets	
	n=25	n=100	n=25	n=100	n=25	n=100	n=25	n=100
Random sampling	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Difficulty-stratified sampling	0.88 (0.81–0.97)	0.77 (0.69–0.86)	0.82 (0.75–0.90)	0.81 (0.74–0.88)	0.86 (0.80–0.93)	0.81 (0.74–0.90)	0.92 (0.84–1.00)	0.90 (0.84–0.98)
Anchor points + IRT (tinyBenchmarks-style)	0.96 (0.80–1.19)	1.00 (0.74–1.30)	1.02 (0.85–1.24)	1.37 (1.09–1.74)	1.34 (1.11–1.67)	1.97 (1.61–2.50)	1.36 (0.98–1.87)	2.16 (1.47–3.15)
Random + ridge regression	1.12 (0.95–1.32)	1.24 (1.08–1.47)	1.13 (0.93–1.38)	1.60 (1.33–1.98)	1.43 (1.16–1.81)	2.13 (1.75–2.67)	1.48 (1.02–2.07)	2.09 (1.41–3.05)
Random + IRT-AIPW	0.82 (0.76–0.92)	0.71 (0.66–0.79)	0.78 (0.71–0.87)	0.84 (0.78–0.91)	0.83 (0.75–0.93)	0.82 (0.75–0.89)	0.93 (0.84–1.03)	0.95 (0.88–1.02)
VA-PPS + IRT (pure)	1.00 (0.84–1.22)	1.16 (1.04–1.36)	1.03 (0.87–1.25)	1.48 (1.28–1.77)	1.33 (1.11–1.62)	2.07 (1.77–2.49)	1.38 (1.00–1.84)	2.12 (1.50–2.99)
VA-PPS + Cal-IRT (ours)	0.72 (0.63–0.82)	0.68 (0.61–0.76)	0.71 (0.62–0.81)	0.73 (0.67–0.81)	0.84 (0.76–0.95)	0.82 (0.71–0.95)	0.90 (0.77–1.06)	0.92 (0.81–1.05)
VA-PPS + ShrinkCal-IRT (ours)	0.83 (0.72–0.98)	0.80 (0.73–0.91)	0.81 (0.71–0.94)	0.85 (0.79–0.93)	0.95 (0.84–1.09)	0.93 (0.82–1.07)	1.01 (0.84–1.20)	1.07 (0.93–1.24)

Answer: [Yes]

Justification: Proposition 1 states its assumptions and has a short complete proof; the heuristics (Cal-IRT, ShrinkCal-IRT, VA-PPS proxy) are explicitly presented without guarantees.

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Table 16: Mean signed error (accuracy points, averaged over benchmarks) and bias^2/MSE at $n = 100$ under different shifts.

	Top-10% targets		Stronger targets (extrapolation)		Top-50% targets		Weakest-25% targets	
	mean err (pts)	bias^2/MSE	mean err (pts)	bias^2/MSE	mean err (pts)	bias^2/MSE	mean err (pts)	bias^2/MSE
Random sampling	-0.07	55%	-0.55	41%	-0.07	45%	+0.04	26%
Difficulty-stratified sampling	+0.10	26%	-0.15	18%	-0.12	16%	-0.13	11%
Anchor points + IRT (tinyBenchmarks-style)	-3.81	74%	-5.01	72%	-7.51	70%	+9.01	70%
Random + ridge regression	-4.43	81%	-5.58	76%	-7.99	73%	+8.15	67%
Random + IRT-AIPW	-0.20	17%	-0.33	19%	-0.31	21%	+0.21	22%
VA-PPS + IRT (pure)	-4.08	78%	-5.17	76%	-7.58	73%	+8.23	69%
VA-PPS + Cal-IRT (ours)	-0.63	25%	-0.41	16%	-0.74	24%	+1.31	25%
VA-PPS + ShrinkCal-IRT (ours)	-1.62	40%	-1.44	29%	-1.79	35%	+2.28	40%

Table 17: Relative RMSE vs. number of source models m, $n = 25$.

	Random split				Stronger targets			
	m=20	m=50	m=100	m=all	m=20	m=50	m=100	m=all
Difficulty-stratified sampling	0.82 (0.74–0.91)	0.84 (0.75–0.94)	0.82 (0.74–0.91)	0.80 (0.72–0.90)	0.79 (0.70–0.89)	0.80 (0.68–0.91)	0.79 (0.69–0.90)	0.79 (0.70–0.89)
VA-PPS + ridge regression	0.71 (0.52–0.95)	0.56 (0.42–0.74)	0.52 (0.38–0.70)	0.51 (0.37–0.69)	1.22 (0.88–1.70)	1.05 (0.75–1.47)	0.97 (0.70–1.30)	0.96 (0.70–1.30)
VA-PPS + IRT (pure)	0.77 (0.55–1.02)	0.61 (0.46–0.79)	0.55 (0.43–0.72)	0.56 (0.43–0.72)	1.47 (1.04–2.06)	1.14 (0.83–1.55)	1.01 (0.75–1.31)	0.87 (0.69–1.12)
VA-PPS + IRT-AIPW	0.96 (0.80–1.15)	0.82 (0.70–0.97)	0.71 (0.62–0.82)	0.73 (0.63–0.84)	1.51 (0.92–2.06)	0.97 (0.66–1.37)	0.84 (0.63–1.11)	0.79 (0.65–0.96)
VA-PPS + Cal-IRT	0.71 (0.58–0.86)	0.65 (0.54–0.77)	0.63 (0.53–0.74)	0.64 (0.54–0.75)	0.86 (0.67–1.07)	0.76 (0.61–0.95)	0.74 (0.60–0.91)	0.68 (0.56–0.81)
VA-PPS + ShrinkCal-IRT	0.70 (0.54–0.88)	0.62 (0.50–0.75)	0.57 (0.46–0.69)	0.58 (0.47–0.71)	0.98 (0.77–1.22)	0.87 (0.67–1.10)	0.81 (0.64–0.98)	0.73 (0.59–0.91)

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Table 18: Relative RMSE vs. number of source models m, $n = 100$.

	Random split				Stronger targets			
	m=20	m=50	m=100	m=all	m=20	m=50	m=100	m=all
Difficulty-stratified sampling	0.86 (0.78–0.96)	0.83 (0.76–0.91)	0.85 (0.77–0.93)	0.83 (0.75–0.91)	0.83 (0.73–0.92)	0.82 (0.74–0.90)	0.84 (0.77–0.93)	0.83 (0.74–0.92)
VA-PPS + ridge regression	0.92 (0.75–1.10)	0.75 (0.62–0.89)	0.75 (0.61–0.91)	0.66 (0.53–0.82)	1.96 (1.43–2.81)	1.71 (1.30–2.30)	1.60 (1.25–2.12)	1.42 (1.10–1.83)
VA-PPS + IRT (pure)	1.25 (1.00–1.49)	0.86 (0.72–1.01)	0.80 (0.66–0.95)	0.71 (0.61–0.83)	2.63 (1.98–3.70)	1.96 (1.48–2.62)	1.70 (1.34–2.18)	1.40 (1.20–1.60)
VA-PPS + IRT-AIPW	1.12 (1.00–1.24)	0.85 (0.73–1.00)	0.83 (0.74–0.93)	0.78 (0.69–0.89)	1.16 (0.92–1.51)	0.96 (0.74–1.24)	0.87 (0.73–1.04)	0.82 (0.69–0.96)
VA-PPS + Cal-IRT	0.84 (0.71–0.95)	0.75 (0.65–0.84)	0.76 (0.67–0.85)	0.73 (0.65–0.82)	0.94 (0.75–1.22)	0.85 (0.70–1.04)	0.81 (0.65–1.00)	0.73 (0.63–0.83)
VA-PPS + ShrinkCal-IRT	0.86 (0.74–0.97)	0.71 (0.61–0.82)	0.72 (0.62–0.84)	0.68 (0.60–0.77)	1.10 (0.92–1.34)	0.95 (0.83–1.12)	0.90 (0.75–1.07)	0.82 (0.73–0.93)

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